Enhanced Adaptive Higher Order Sliding Mode Observer based Sensorless Control

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***Abstract*— The traditional higher order sliding mode observer (HSMO) based sensorless control strategy uses a constant gain. However, very large gain value can decrease the output accuracy, while very small gain value will degrade the sliding mode observer's stability. Therefore, under uncertain disturbances, an adaptive gain should be set for the higher order sliding mode observer. This paper proposes a variable universe fuzzy adaptive high order sliding mode observer based sensorless control (VUF-HSMO) strategy for permanent magnet synchronous machines (PMSMs) to achieve high precision sensorless control. Firstly, fuzzy logic control technique is used to get an adaptive gain in the HSMO. Then, variable universe theory is employed to self-tune the universe of discourse according to the change of inputs, and this results in the improvement in position estimation accuracy. Experiment results under different conditions demonstrate the effectiveness and the superiority of the proposed VUF-HSMO based sensorless control compared with the traditional HSMO and fuzzy logic based HSMO.**

Keywords—sensorless control, sliding mode observer, PMSMs

I. INTRODUCTION

Permanent magnet synchronous machines (PMSMs) are widely used in electric vehicles (EVs), due to their high efficiency, high power density and compact size [1]-[3]. For PMSMs, high-precision control strategies, such as maximum torque control and maximum efficiency control, are crucial for superior system performance, which rely on sensors to obtain the accurate information of rotor position and speed. However, mechanical sensors are expensive, bulky and susceptible to environment [4]-[5]. Therefore, there has been an increasing attention paid to the sensorless control strategy of PMSMs. Moreover, such strategies can also serve as a substitute during sensor malfunction.

In recent years, model-based methods have been considered as effective sensorless control strategies for PMSMs in the medium- and high-speed regions due to its non-invasiveness [6]. Among model-based methods, sliding mode observer (SMO) based method is widely used due to its strong robustness, insensitivity to parameter variation, anti-interference ability, and good dynamic performance [7-8]. However, the chattering problem, caused by the discontinuous switching characteristics of sliding-mode variable structure control method, is an inherent defect of SMO [9]. This chattering phenomenon becomes more serious in the low speed region, which limits the working range for SMO based sensorless control method. Therefore, many switching functions have been proposed to replace the discrete switching control function in the traditional SMO, such as saturation function and sigmoid function to obtain a trade-off between chattering reduction and system robustness [10-13] . In [14-15], the super-twisting algorithm (STA), which is a well-known second order sliding mode algorithm, is introduced in the SMO based sensorless control to eliminate chattering. To further enhance the anti-chattering and estimation performance, high order sliding mode observer (HOSMO) is proposed in [15]. Based on the comparative study conducted in [16], the precision of the sliding manifold is improved by using the HOSMO rather than STA based SMO and the robustness to chattering effect has also been improved. However, the selection of sliding mode gains and boundary layer are difficult in the design of HOSMO. Too large gain can decrease the output accuracy, while too small gain will destroy the sliding mode observer's stability.

Artificial intelligence based sensorless control technique by using fuzzy logic has been implemented for PMSM application in recent years, which can minimize the tuning efforts and improve the estimated response without requiring the mathematical model of motor [17]. Therefore, fuzzy logic control theory is employed in the SMO based sensorless control strategy to adapt the gain to suppress the chattering phenomenon without decreasing system stability [18]-[19]. However, the control rules in fuzzy logic controller is usually designed based on experts' experiences or trial and error methods [20]. The limited number of fuzzy control rules will reduce the control precision, while too many fuzzy control rules will increase the difficulty of the design procedure and lead to the so-called rules explosion problem. To maintain the high control accuracy without redundant control rules, the variable universe fuzzy adaptive high order sliding mode observer (VUF-HSMO) based sensorless control is proposed in this paper to attenuate the chattering effect. By self-tuning of the universe of discourse, the inputs and outputs will change, which allows the observer to select different fuzzy control rules without the changes of the rule bases. Comparative results are provided in this paper to demonstrate the effectiveness and the satisfactory chattering reduction of the proposed VUF-HSMO.

The remainder of this paper is organized as follows. The traditional HSMO is reviewed in Section II. Thereafter, Section III presents the design of the proposed VUF-HSMO. Experiment results are given in Section IV. Finally, some concluding remarks are presented in Section V.

II. CONVENTIONAL HSMO

The design of the conventional HSMO algorithm is based on the motor mathematical model in the αβ-axis system.

$$\begin{cases} u_{\alpha\beta} = \begin{bmatrix} R_s + pL_d & \omega_e(L_d - L_q) \\ -\omega_e(L_d - L_q) & R_s + pL_d \end{bmatrix} i_{\alpha\beta} + e_{\alpha\beta} \\ e_{\alpha\beta} = \left[(L_d - L_q)(\omega_e i_d - p i_q) + \omega_e \psi_f \right] \begin{bmatrix} -\sin\theta_e \\ \cos\theta_e \end{bmatrix} \end{cases} \quad (1)$$

where $u_{\alpha\beta} = \begin{bmatrix} u_\alpha & u_\beta \end{bmatrix}^T$ and $i_{\alpha\beta} = \begin{bmatrix} i_\alpha & i_\beta \end{bmatrix}^T$ are the stator voltage and current vectors in the α-β axes, respectively;

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$e_{\alpha\beta} = \begin{bmatrix} e_\alpha & e_\beta \end{bmatrix}^T$ is the extended back electromotive force (EEMF) in the α-β axes; R_s is the stator resistance of PMSM; L_d and L_q are the d–q axis inductances of PMSM; ω_e is the rotor electric angular speed; θ_e is the electrical angle of the rotor; ψ_f is the permanent magnet flux, and p = d/dt is the differential operator. The EEMF contains all the information of the rotor position and speed of the motor. Therefore, by accurately obtaining the EEMF, the speed and position information of the motor can be calculated.

During one sampling period, the rotor speed can be considered as a constant $(\dot{\omega}_e \approx 0)$. Hence, the EEMF dynamics can be expressed as (2).

$$\begin{aligned} \begin{bmatrix} \dot{e}_\alpha \\ \dot{e}_\beta \end{bmatrix} &= \begin{bmatrix} (L_d - L_q)(\omega_e i_d - p i_q) + \omega_e \psi_f \end{bmatrix} \begin{bmatrix} -\omega_e \cos\theta_e \\ -\omega_e \sin\theta_e \end{bmatrix} \\ &= \omega_e \begin{bmatrix} -e_\beta \\ e_\alpha \end{bmatrix} \end{aligned} \tag{2}$$

Based on (1) and (2), the high order state equation with current and EEMF as state variables can be derived as follows:

$$\begin{bmatrix} \dot{i}_{\alpha\beta} \\ \dot{e}_{\alpha\beta} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} i_{\alpha\beta} \\ e_{\alpha\beta} \end{bmatrix} + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} u_{\alpha\beta} \tag{3}$$

with

$$A_{11} = \begin{bmatrix} -\frac{R_s}{L_d} & -\frac{\omega_e (L_d - L_q)}{L_d} \\ \frac{\omega_e (L_d - L_q)}{L_d} & -\frac{R_s}{L_d} \end{bmatrix}, \; A_{12} = \begin{bmatrix} -\frac{1}{L_d} & 0 \\ 0 & -\frac{1}{L_d} \end{bmatrix},$$

$$A_{22} = \begin{bmatrix} 0 & -\omega_e \\ \omega_e & 0 \end{bmatrix}, \; \mathrm{B}_1 = \begin{bmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_d} \end{bmatrix}.$$

To obtain the estimated EEMF, the HSMO is designed as (4).

$$\begin{bmatrix} \dot{\hat{i}}_{\alpha\beta} \\ \dot{\hat{e}}_{\alpha\beta} \end{bmatrix} = \begin{bmatrix} \hat{A}_{11} & A_{12} \\ 0 & \hat{A}_{22} \end{bmatrix} \begin{bmatrix} \hat{i}_{\alpha\beta} \\ \hat{e}_{\alpha\beta} \end{bmatrix} + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} u_{\alpha\beta} - \frac{1}{L_d} K sign(\hat{i}_{\alpha\beta} - i_{\alpha\beta}) \tag{4}$$

where $K = \begin{bmatrix} k_1 & 0 & k_2 & 0 \\ 0 & k_1 & 0 & k_2 \end{bmatrix}^T$ is the gain matrix for HSMO designed according to the Lyapunov stability principle and the symbol "^" represents the estimated value.

Subtracting (3) from (4), the error dynamics of the current and EEMF estimation can be obtained as (5).

$$\begin{cases} \begin{bmatrix} \dot{\tilde{i}}_{\alpha\beta} \end{bmatrix} = \frac{(L_d - L_q)}{L_d} \begin{bmatrix} 0 & -\tilde{\omega}_e \\ \tilde{\omega}_e & 0 \end{bmatrix} \tilde{i}_{\alpha\beta} - \frac{k_1}{L_d} sign(\tilde{i}_{\alpha\beta}) - \frac{1}{L_d} \tilde{e}_{\alpha\beta} \\ \begin{bmatrix} \dot{\tilde{e}}_{\alpha\beta} \end{bmatrix} = \begin{bmatrix} 0 & -\tilde{\omega}_e \\ \tilde{\omega}_e & 0 \end{bmatrix} \tilde{e}_{\alpha\beta} + k_2 sign(\tilde{i}_{\alpha\beta}) \end{cases} \tag{5}$$

where, the symbol "~" represents the error dynamics. Based on [15], through tuning the parameters k_1 and k_2, the difference between the EEMF and the real ones can settle to the desired vicinity. According to the equivalent control principle of sliding mode control, the control quantity at this time can be regarded as the equivalent control quantity, that is, the estimated EEMF is equal to the actual EEMF. Thereafter, the HSMO can estimate the rotor position and speed through the estimated EEMF.

The estimated EEMF can be expressed as (6) by using Laplace to (5).

$$\begin{bmatrix} \tilde{e}_\alpha \\ \tilde{e}_\beta \end{bmatrix} = \frac{k_2}{s^2 + \tilde{\omega}_e^2} \begin{bmatrix} s & -\tilde{\omega}_e \\ \tilde{\omega}_e & s \end{bmatrix} sign(\tilde{i}_{\alpha\beta}) \tag{6}$$

The estimated EEMF is filtered out since $k_2 / \left(s^2 + \tilde{\omega}_e^2\right)$ behaves as a second order low pass filter (LPF). Therefore, there is no need to design extra LPF in the HSMO, which avoids the phase delay problem compared with traditional first order SMO. Thereafter, the HSMO can estimate the rotor position and speed through the estimated EEMF.

The phase locked loop (PLL) is usually used to extract the position information from the EEMF due to its characteristic of low-pass filter. Block diagram of the conventional PLL is shown in Fig.1. Let η denote the $(L_d - L_q)(\omega_e i_d - p i_q) + \hat{\omega}_e \psi_f$ term, which is the magnitude of the EEMF, (7) can be obtained based on (1), if $(\theta_e - \hat{\theta}_e)$ is very small.

$$\begin{aligned} \Delta e &= -\hat{e}_\alpha \cos\hat{\theta}_e - \hat{e}_\beta \sin\hat{\theta}_e = \eta(\sin\theta_e \cos\hat{\theta}_e - \cos\theta_e \sin\hat{\theta}_e) \\ &= \eta \sin(\theta_e - \hat{\theta}_e) \approx \eta(\theta_e - \hat{\theta}_e) \end{aligned} \tag{7}$$

Then, the position transfer function can be obtained [15].

$$G(s) = \frac{\hat{\theta}_e}{\theta_e} = \frac{2\xi\omega_n s + \omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \tag{8}$$

with $\xi = \sqrt{\eta K_i}$ and $\omega_n = \frac{K_p}{2}\sqrt{\frac{\eta}{K_i}}$

where K_p is the proportional gain and K_i is the integral gain.

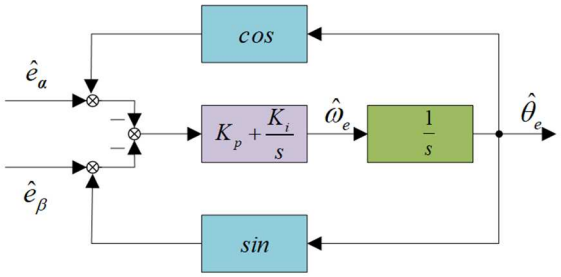


Fig. 1. Traditional PLL block.

Despite the fact that the HSMO is effective at reducing chattering, there is still a problem to be overcome. While using constant gains in this method, the gains should be large enough to meet the stable conditions when the motor is running at high speed. However, due to the high sliding mode gains, the performance of the HSMO in the low speed region will be deteriorated.

III. Variable Universe Fuzzy Adaptive HSMO

According to the analysis above, the gain values in the conventional HSMO are constant values and need to be set large to ensure the robustness of the HSMO when EEMF is

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high. However, the observed current error $\tilde{i}_{\alpha\beta}$ in the HSMO changes dynamically. When $|\tilde{i}_{\alpha\beta}|$ is large, the system is far away from the sliding surface, a corresponding large gain should be set to reduce the response time. When $|\tilde{i}_{\alpha\beta}|$ is small, the system is close to the sliding surface, a corresponding small gain should be set to avoid the overshoot of the sliding motion. Therefore, an appropriate gain should be regulated to perform accurate position estimation and eliminate the chattering problems.

To adjust the observer gain adaptively in SMO, the fuzzy logic control is firstly employed [18]. The universe of discourse of the input signals are chosen as the observed current error "s" and its derivative "ṡ". The controller employs triangle membership function in this paper. The fuzzy rule base with 49 rules is illustrated in Table I. The NB, NM, NS, ZO, PS, PM, PB represent the negative big, the negative medium, the negative small, zero, the positive small, the positive medium and the positive big, respectively.

TABLE I. FUZZY CONTROL RULES

ṡ \ s	NB	NM	NS	ZO	PS	PM	PB
NB	PB	PB	PB	PB	PM	PS	ZO
NM	PB	PB	PB	PM	PS	ZO	NS
NS	PB	PB	PM	PS	ZO	NS	NM
ZO	PB	PM	PS	ZO	NS	NM	NB
PS	PM	PS	ZO	NS	NM	NB	NB
PM	PS	ZO	NS	NM	NB	NB	NB
PB	ZO	NS	NM	NB	NB	NB	NB

However, the fuzzy controller is designed based on experts' experiences or trial and error methods. If the universe range is set too small, its accuracy will reduce and the overall system performance will be destroyed. Conversely, if the universe range is set too large, this will cause the system to be unstable. Therefore, the VUF-HSMO based sensorless control strategy is proposed in this paper. In this method, the input and output universes of discourse can be adjusted properly according to the external operating conditions without the change of the rule base.

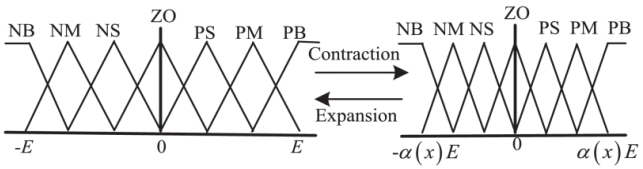


Fig. 2. Illustration of variable universe [22].

The idea of variable universe is first proposed in [21], which indicates that the universe of discourse of the input and output signal changes along with the varying of the input variables. The concept of variable universe is illustrated in Fig. 2 [22]. In Fig. 2, the initial fuzzy domain of input variable x is [-E, E]. By applying variable universe theory into the fuzzy logic control, the fuzzy rules do not change in form while the fuzzy domain changes by multiplying an input contraction-expansion factor α(x) which changes with the error to eliminate control dead zone and increase control accuracy. α(x) can be properly designed as (9) [23].

$$\alpha(x) = \varepsilon + (|x| / E)^{\tau} \tag{9}$$

where ε and τ are parameters for the input contraction-expansion factor. The design of ε and τ can refer to [23]. The stability of the variable universe theory has been proved mathematically and the effectiveness has been demonstrated on various real-time trajectory-tracking nonlinear systems [17]. In this paper, the input variables are the observed current error "s" and its derivative "ṡ". The structure of proposed VUF-HSMO based sensorless control strategy is shown in Fig. 3.

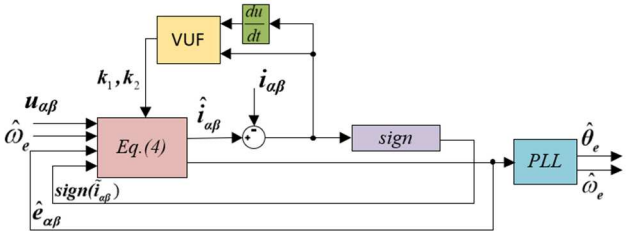


Fig. 3. Block diagram of the overall VUF-HSMO.

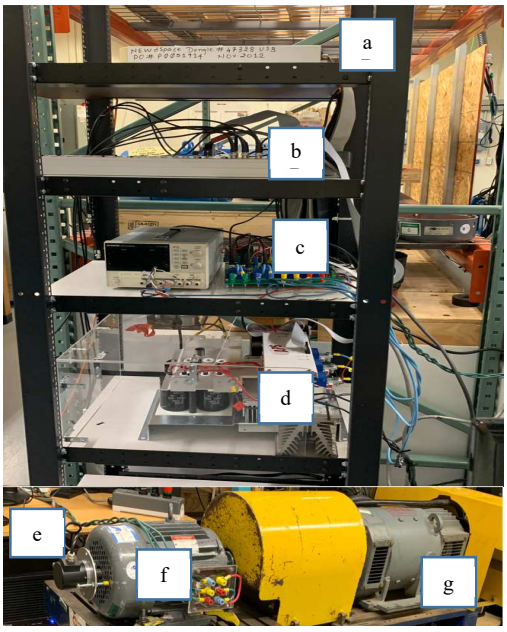


Fig. 4. Experiment setup. (a) dSPACE DC1103 system, (b) dSPACE connector panel, (c) sensor board, (d) inverter, (e) encoder, (f) Surface mounted PMSM, (g) DC dynamometer..

IV. EXPERIMENT RESULTS

The performance of the proposed VUF-HSMO is tested on a VSI-fed laboratory PMSM drive system as shown in Fig. 4. The experimental setup consists of a standard two-level inverter to drive the laboratory PMSM, and the switching frequency is 10 kHz. The PMSM is coupled with a DC dynamometer through the torque transducer setup. The sensor board is designed to measure the three phase motor currents as well as the DC-bus voltage input to the inverter. A 12-bit absolute encoder is used to measure the actual rotor position that is used for motor starting and validation of the estimated position during sensorless control. The proposed VUF-HSMO based sensorless control has been programed into the dSPACE real-time simulator with a control cycle of

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100 μs. To improve the execution speed, the proposed fuzzy logic controller is implemented using a lookup table as shown in Fig.5. The parameters of the PMSM under test are listed in Table II.

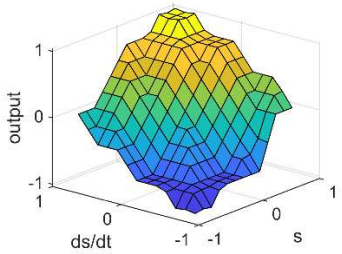


Fig. 5. Fuzzy control surface.

TABLE II. PRAMETERS OF THE PMSM UNDER TEST

Rated Speed	1350 RPM	Rated Torque	25 Nm
Rated current	15 A	Rated voltage	220 V rms
PM flux linkage	0.42 Wb	dq-axis Inductance	0.00335 H
Stator resistance	0.1718 Ω	Number of poles	4
Moment of inertia	0.0384 kg/m^2	Viscous friction	0.000425 Nm.s

The performance of the proposed VUF-HSMO has been tested under both low speed (20 rad/sec) and high speed (100 rad/sec) in comparison with the traditional HSMO and F-HSMO. The motor speed waveforms under FOC with position sensor at 20 rad/sec and 100 rad/sec are shown in Fig.6, where ω_e^* is the reference speed, and ω_e is the measured speed. It is evident that the measured motor speed can follow the reference speed with high accuracy by using the installed position sensor.

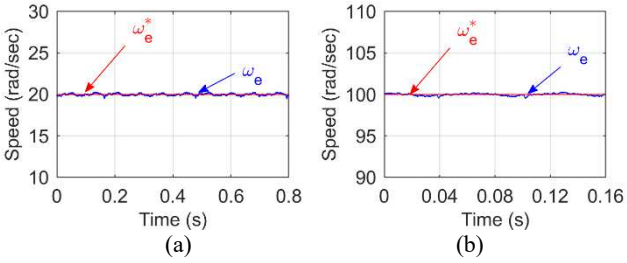


Fig. 6. Speed waveforms with position sensor. (a) 20rad/sec, (b)100rad/sec.

The speed and position waveforms under 20 rad/sec and 100 rad/sec for three different SMO based sensorless strategies are shown in Figs. 7-9 and Figs.10-12, where ω_{smo} and θ_{smo} are the SMO estimated speed and position, respectively; $\Delta\omega$ and $\Delta\theta$ are the estimated speed and position errors, respectively; and θ_e^* is the real motor position measured by the absolute encoder. Compared Fig.9 (a) with Fig.7 (a) and Fig.8 (a), the proposed VUF-HSMO shows a significant chattering reduction under low speed region compared with traditional HSMO and F-HSMO. The position error for the proposed VUF-HSMO is the smallest among these three SMO methods. In the high speed region, the proposed VUF-HSMO is also accurate with the smallest speed ripples and position errors. Therefore, the chattering reduction performance of the proposed VUF-HSMO under both low speed and high speed can be validated.

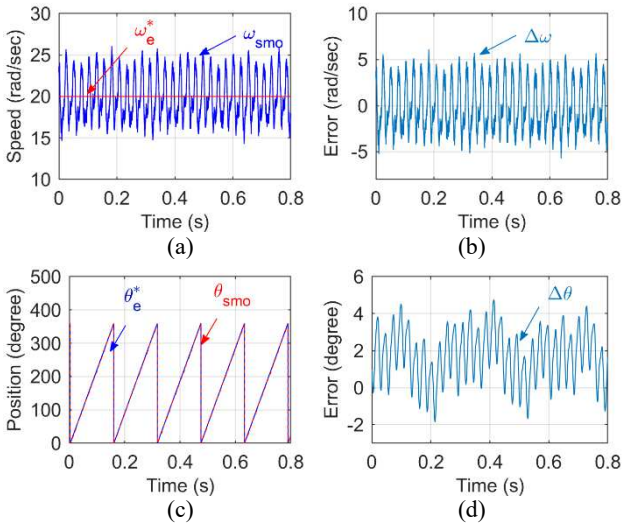


Fig. 7. Traditional HSMO performance under 20 rad/sec. (a) Speed. (b) Speed error. (c) Position. (d) Position error.

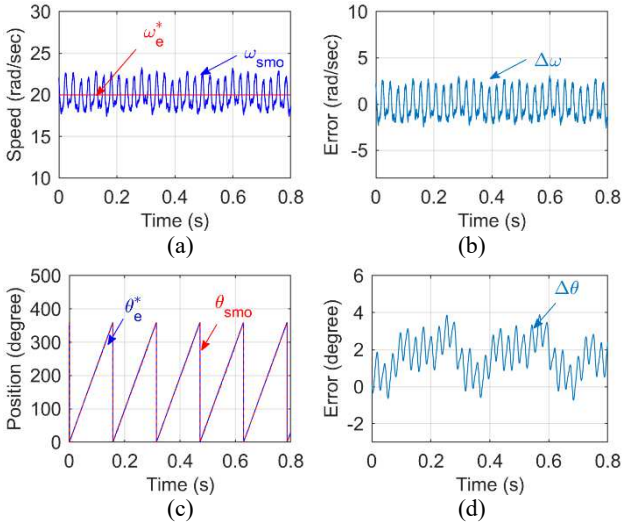


Fig. 8. F-HSMO performance under 20 rad/sec. (a) Speed. (b) Speed error. (c) Position. (d) Position error.

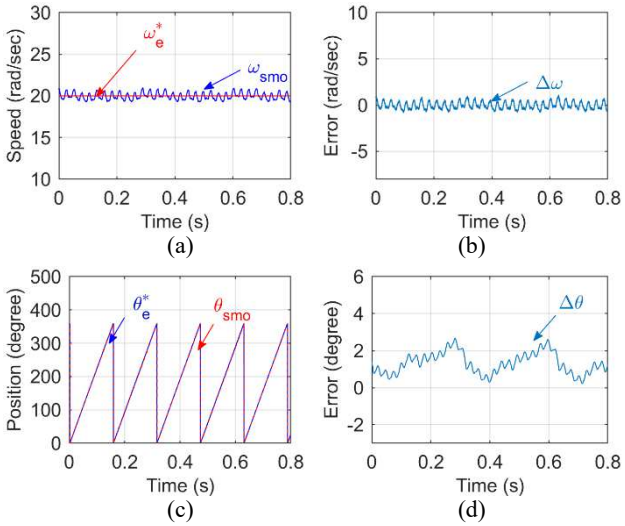


Fig. 9. VUF-HSMO performance under 20 rad/sec. (a) Speed. (b) Speed error. (c) Position. (d) Position error.

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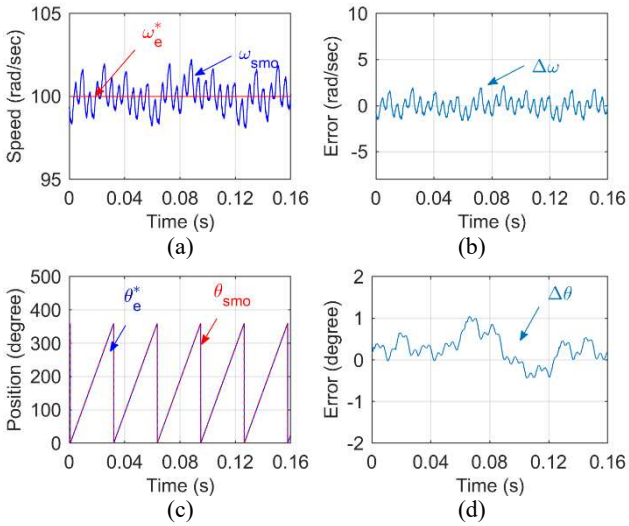


Fig. 10. Traditional-HSMO performance under 100 rad/sec. (a) Speed. (b) Speed error. (c) Position. (d) Position error.

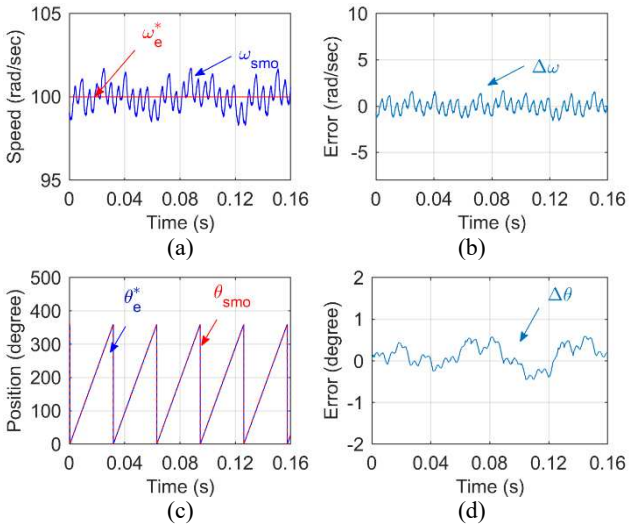


Fig. 11. F-HSMO performance under 100 rad/sec. (a) Speed. (b) Speed error. (c) Position. (d) Position error.

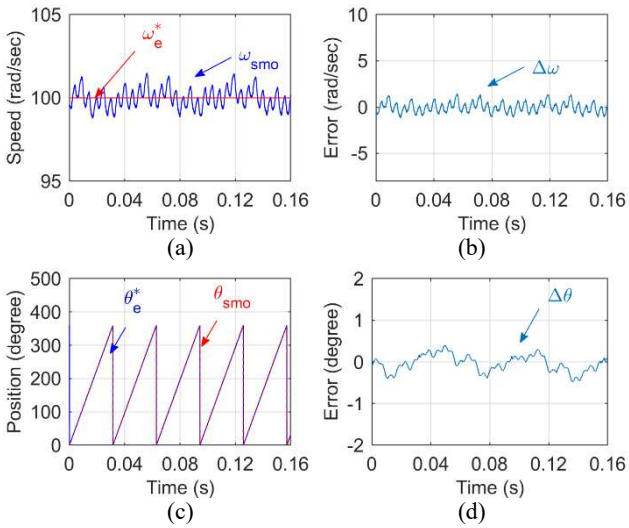


Fig. 12. VUF-HSMO performance under 100 rad/sec. (a) Speed. (b) Speed error. (c) Position. (d) Position error.

V. Conclusion

The variable universe fuzzy adaptive high order sliding mode observer based sensorless control strategy for PMSM is presented in this paper. A self-tuning algorithm based on variable universe theory combined with the fuzzy logic system is employed for online tuning of the sliding mode gains, which has good chattering reduction along with a high estimation accuracy. Experimental results have proven that the proposed method improves the position tracking performance with smaller control action chattering in comparison with the traditional HSMO and F-HSMO, and it leads to a smaller speed ripple during sensorless operation. In order to further reduce algorithm complexity while ensuring accuracy, another lookup table based fuzzy logic controller can be added in the future work to replace the determination process of the contraction–expansion factors. In addition, the inverter nonlinearity and external disturbance compensations, especially in the low speed region, needs to be further investigated to improve the proposed observer performance.

References

[1] G. Wang, R. Yang and D. Xu, "DSP-Based Control of Sensorless IPMSM Drives for Wide-Speed-Range Operation," *IEEE Trans. Ind. Electron.*, vol. 60, no. 2, pp. 720-727, Feb. 2013.

[2] S. N. Afrasiabi and C. Lai, "Investigation of LC Filter Unbalance in an Inverter-Fed Permanent Magnet Synchronous Machine Drive," *2020 23rd International Conference on Electrical Machines and Systems (ICEMS)*, 2020, pp. 1961-1966.

[3] P. Pillay and R. Krishnan, "Application characteristics of permanent magnet synchronous and brushless dc motor for servo drive," *IEEE Trans. Ind. Appl.*, vol. 27, no. 5, pp. 986–996, Sep./Oct. 1991.

[4] F. Parasiliti, R. Petrella, and M. Tursini, "Sensorless speed control of a PM synchronous motor by sliding mode observer," in *Proc. IEEE ISIE*, Jul. 1997, vol. 3, pp. 1106–1111.

[5] L. Ding, Y. W. Li and N. R. Zargari, "Discrete-Time SMO Sensorless Control of Current Source Converter-Fed PMSM Drives With Low Switching Frequency," *IEEE Trans. Ind. Electron.*, vol. 68, no. 3, pp. 2120-2129, March 2021.

[6] G. Wang, M. Valla and J. Solsona, "Position Sensorless Permanent Magnet Synchronous Machine Drives—A Review," *IEEE Trans. Power Electron.,* vol. 67, no. 7, pp. 5830-5842, July 2020.

[7] Y. Wang, Y. Xu and J. Zou, "Sliding-Mode Sensorless Control of PMSM With Inverter Nonlinearity Compensation," IEEE Trans. Power Electron., vol. 34, no. 10, pp. 10206-10220, Oct. 2019.

[8] D. Liang, J. Li, R. Qu and W. Kong, "Adaptive Second-Order Sliding-Mode Observer for PMSM Sensorless Control Considering VSI Nonlinearity," IEEE Trans. Power Electron., vol. 33, no. 10, pp. 8994-9004, Oct. 2018.

[9] H. Kim, J. Son and J. Lee, "A High-Speed Sliding-Mode Observer for the Sensorless Speed Control of a PMSM," *IEEE Trans. Ind. Electron.*, vol. 58, no. 9, pp. 4069-4077, Sept. 2011.

[10] S. Chi, Z. Zhang and L. Xu, "Sliding-Mode Sensorless Control of Direct-Drive PM Synchronous Motors for Washing Machine Applications," *IEEE Trans. Ind. Appl.*, vol. 45, no. 2, pp. 582-590, Mar./ Apr. 2009.

[11] X. Zhang, "Sensorless induction motor drive using indirect vector controller and sliding-mode observer for electric vehicles," *IEEE Trans. Veh. Technol.,* vol. 62, no. 7, pp. 3010–3018, Sep. 2013.

[12] L. Sheng, W. Li, Y. Wang, M. Fan and X. Yang, "Sensorless Control of a Shearer Short-Range Cutting Interior Permanent Magnet Synchronous Motor Based on a New Sliding Mode Observer," *IEEE Access,* vol. 5, pp. 18439-18450, 2017.

[13] H. Wang, X. Ge and Y. Liu, "Second-Order Sliding-Mode MRAS Observer-Based Sensorless Vector Control of Linear Induction Motor Drives for Medium-Low Speed Maglev Applications," *IEEE Trans. Ind. Electron.*, vol. 65, no. 12, pp. 9938-9952, Dec. 2018.

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[14] Z. Li, S. Zhou, Y. Xiao and L. Wang, "Sensorless Vector Control of Permanent Magnet Synchronous Linear Motor Based on Self-Adaptive Super-Twisting Sliding Mode Controller," *IEEE Access*, vol. 7, pp. 44998-45011, 2019.

[15] G. Wang, Z. Li, G. Zhang, Y. Yu and D. Xu, "Quadrature PLL-Based High-Order Sliding-Mode Observer for IPMSM Sensorless Control With Online MTPA Control Strategy," *IEEE Trans. Energy Convers.,* vol. 28, no. 1, pp. 214-224, March 2013.

[16] A. Chalanga, S. Kamal, L. M. Fridman, B. Bandyopadhyay and J. A. Moreno, "Implementation of Super-Twisting Control: Super-Twisting and Higher Order Sliding-Mode Observer-Based Approaches," *IEEE Trans. Ind. Electron.*, vol. 63, no. 6, pp. 3677-3685, June 2016.

[17] Y. Ahmad Tarmizi, K. Abdul Karim, S. A. Ahmad Tarusan, and A. Jidin, "Review and comparison OF Sensorless techniques to estimate the position and speed of pmsm," *International Journal of Power Electronics and Drive Systems (IJPEDS)*, vol. 8, no. 3, p. 1062, 2017.

[18] S. Ye, "Fuzzy sliding mode observer with dual SOGI-FLL in sensorless control of PMSM drives," *ISA Trans.,* vol. 85, pp. 161–176, Feb. 2019.

[19] C. Wang and D. Cao, "New Sensorless Speed Control of a Hybrid Stepper Motor Based on Fuzzy Sliding Mode Observer," *Energies*, vol. 13, no. 18, p. 4939, Sep. 2020.

[20] H.Pan and D.Jin, "Design and Application of Variable Universe Fuzzy Controller Based on Cat Swarm Optimization", *Mathematical Problems in Engineering*, vol. 2016, 6 pages, 2016.

[21] H. Li, "Variable Universe Adaptive Fuzzy Controller," *Science in China*, vol. 29, pp. 32, Jan. 1999.

[22] Y. Ahmad Tarmizi, K. Abdul Karim, S. A. Ahmad Tarusan, and A. Jidin, "Review and comparison OF Sensorless techniques to estimate the position and speed of pmsm," *International Journal of Power Electronics and Drive Systems (IJPEDS)*, vol. 8, no. 3, p. 1062, 2017.

[23] S. Aziz, H. Wang, Y. Liu, J. Peng and H. Jiang, "Variable Universe Fuzzy Logic-Based Hybrid LFC Control With Real-Time Implementation," *IEEE Access*, vol. 7, pp. 25535-25546, 2019.

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